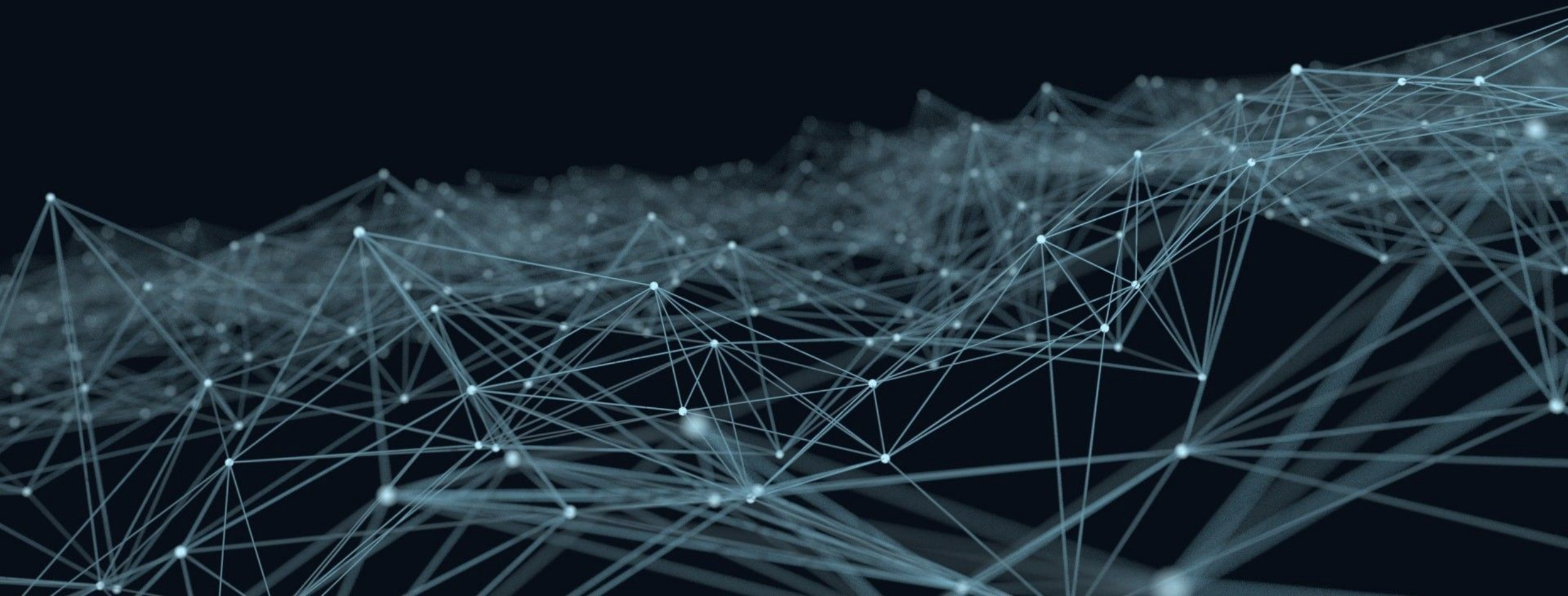
Java Full Stack Development

1. Scanning



TEKsystems Global Services

WORKFORCE DEVELOPMENT

Module Topics

- Static Security Application Testing
- Dynamic Security Application Testing

Static Security Application Testing

SAST

- Static Application Security Testing
  - Static
    - Analyzes source code or compiled bytecode without executing it
  - Application
    - Focuses on the code your team writes, not the OS or infrastructure
  - Security Testing
    - Looks specifically for security defects, not bugs in general
  - Runs early and often
    - Every commit, every pull request, every CI build
    - The "shift left" philosophy
    - Find vulnerabilities at the cheapest point in the SDLC

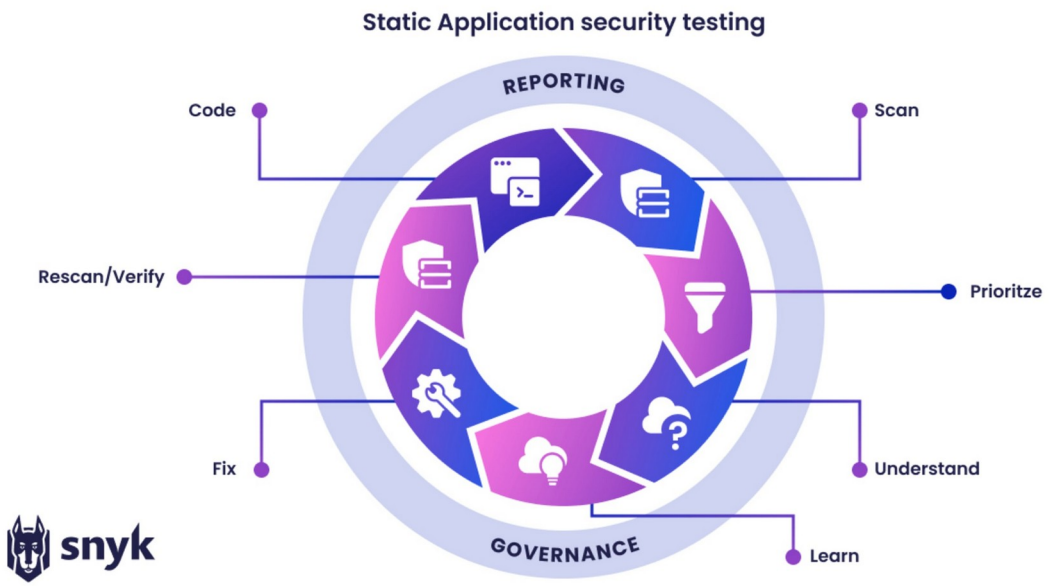


Image Credit:https://snyk.io/fr/articles/application-security/static-application-security-testing/

SAST

- Static
  - Code is analyzed without being runC
  - The analyzer reads the source files
    - Builds a model of what the code does
    - Looks for patterns that indicate vulnerabilities
    - Compare with dynamic analysis (DAST), which runs the application and probes it from outside
    - Static is faster, cheaper, and earlier in the SDLC but dynamic catches things static cannot
- Application
  - For application code
    - Not for operating system vulnerabilities, network misconfigurations, or hardware issues
    - Those are covered by other tools

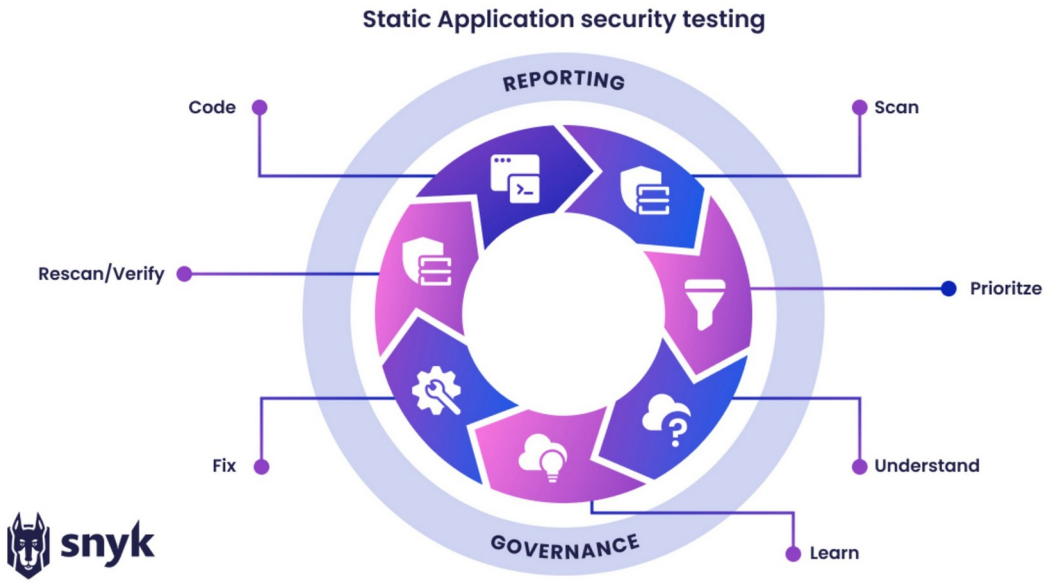
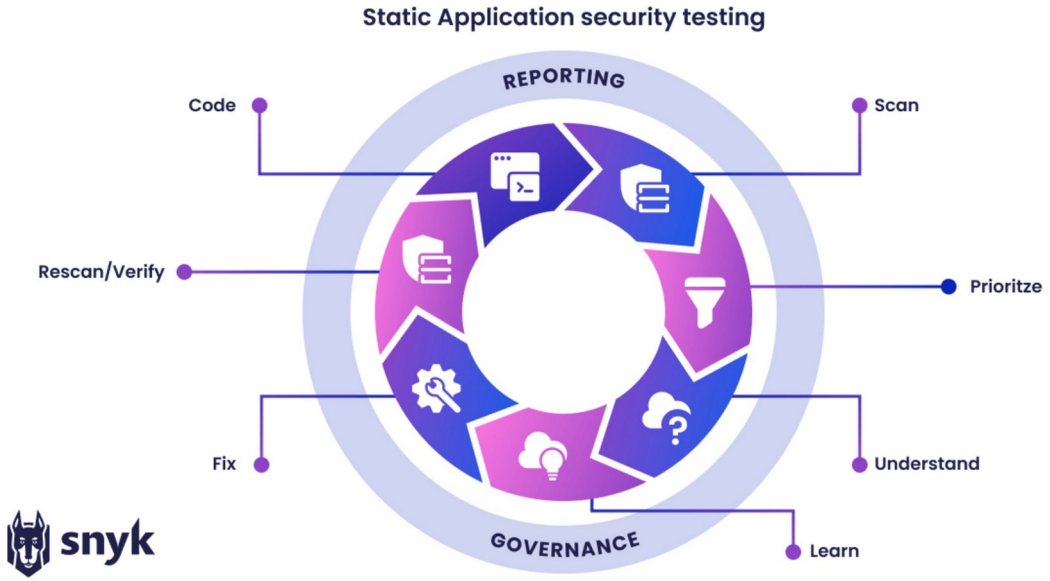


Image Credit:https://snyk.io/fr/articles/application-security/static-application-security-testing/

SAST

- Security testing focus
  - SAST tools are tuned to find security defects
    - SQL injection, XSS, hardcoded credentials, weak crypto
    - Some tools also report code quality issues alongside security issues
  - SonarQube is a SAST too that
    - Does security scans
    - Identifies potential bugs
    - Identifies code smells
    - Provides metrics like cyclomatic complexity
  - Checkmarx is used by enterprises
    - More thorough but requires a licence



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SAST

- How SAST works
  - Step 1: Parse
    - Read source code, build an Abstract Syntax Tree (AST)
  - Step 2: Model
    - Build a control flow graph showing how execution moves through the code
  - Step 3: Analyze
    - Perform taint analysis, tracking untrusted data from sources to sinks
  - Step 4: Report
    - Generate findings for code patterns that match known vulnerability rules

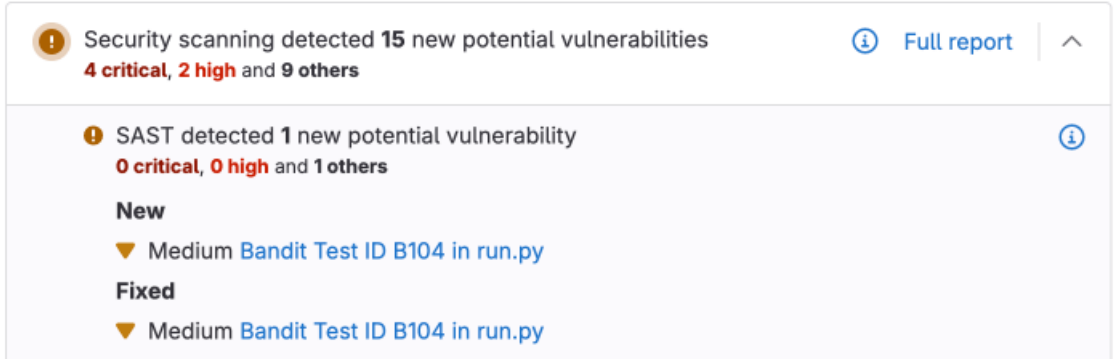
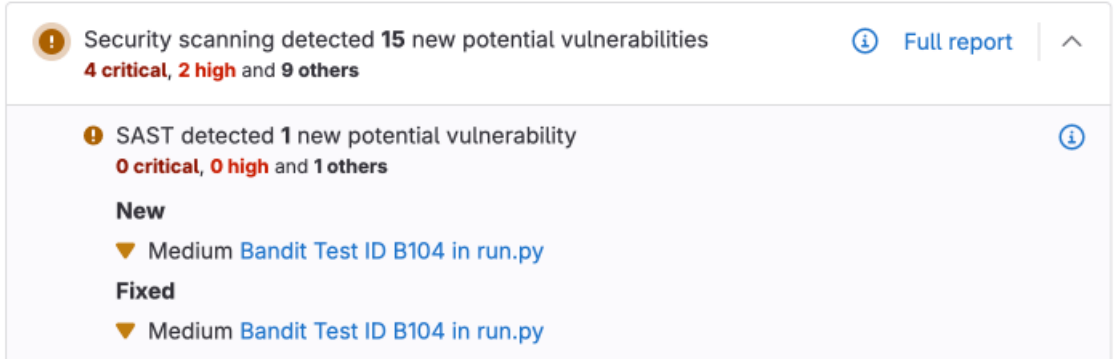


Image Credit: https://gitlab-docs-d6a9bb.gitlab.io/ee/user/application_security/sast/

SAST

- What SAST catches
  - Injection vulnerabilities
    - SQL injection, command injection, LDAP injection, XPath injection
  - Cross-site scripting (XSS)
    - Unescaped user input in HTML output
  - Hardcoded credentials
    - Passwords, API keys, tokens in source files
  - Weak cryptography
    - MD5/SHA-1 for security purposes, ECB mode, hardcoded keys, weak random number generation



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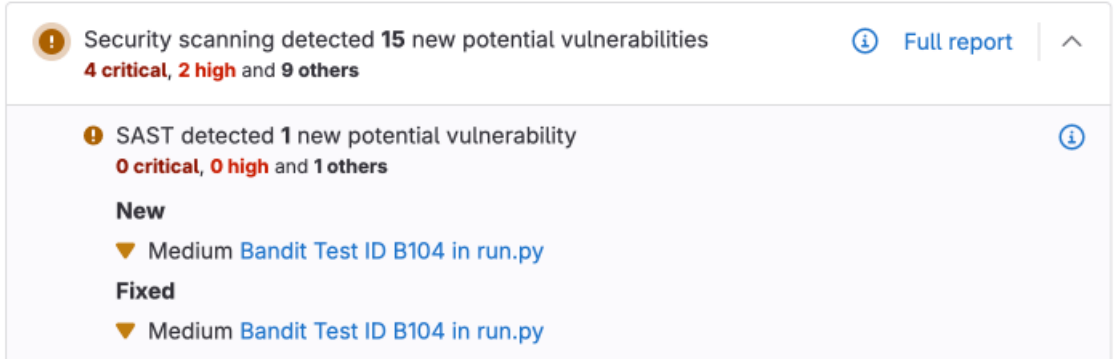
SAST

- What SAST catches
  - Path traversal
    - User input controlling file paths
  - Insecure deserialization
    - Untrusted data passed to readObject(), Jackson, etc.
  - Information leakage
    - Logging sensitive data, exposing stack traces, debug info in responses



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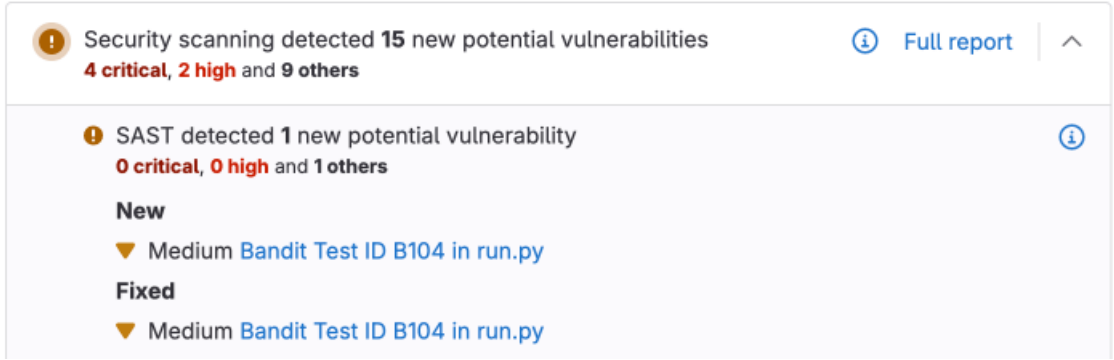
SAST

- Injection
  - The canonical example
  - The pattern is always source-flows-to-sink-without-sanitizer
  - SAST tools detect it well
  - These all share the pattern
    - SQL injection (CWE-89)
    - Command injection (CWE-78)
    - LDAP injection (CWE-90)
    - XPath injection (CWE-91)
    - Header injection (CWE-93)



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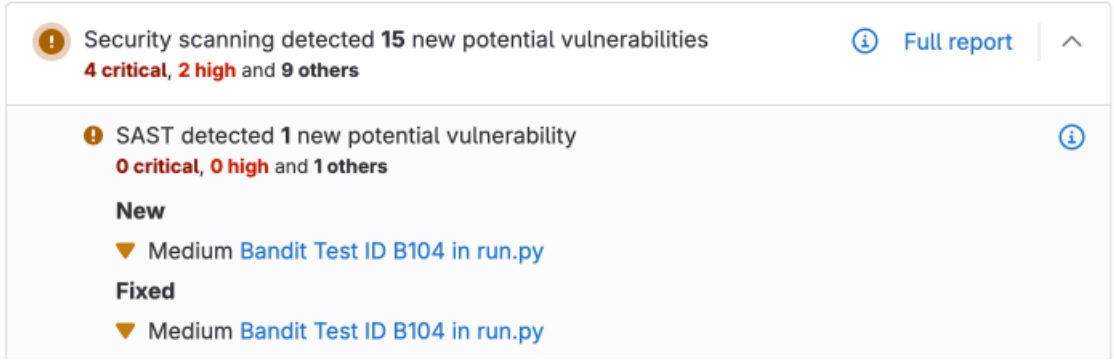
SAST

- Hardcoded credentials
  - Detected by pattern matching
  - Looking for string literals that look like passwords or keys, often with regex against common variable names
  - Less sophisticated than taint analysis but quite effective
- Path traversal
  - Taint analysis applied to file system operations
    - User input flows to new File(...) or Path.of(...)
    - The analyzer asks whether the input was sanitized before use



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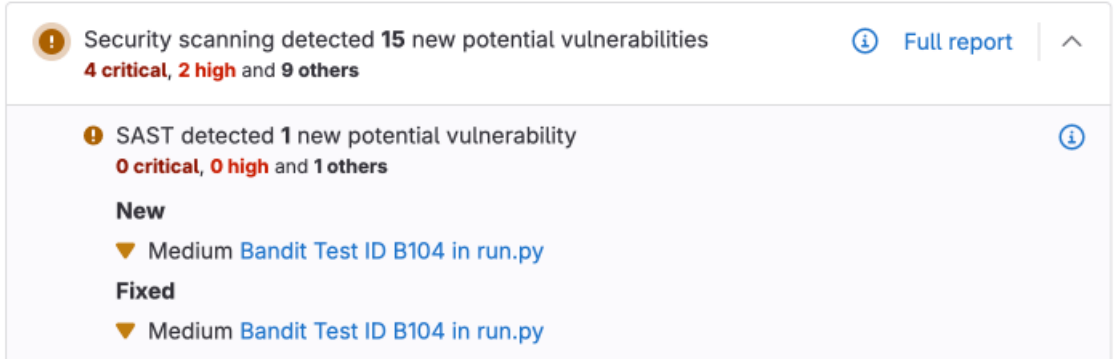
SAST

- What SAST misses
  - Business logic flaws
    - "Users can transfer money from accounts they don't own"
  - Authentication misconfiguration
    - Security rules that aren't applied to the right URLs
  - Authorization bugs
    - Checking the wrong role for the wrong resource
  - Race conditions
    - Concurrency bugs that depend on timing



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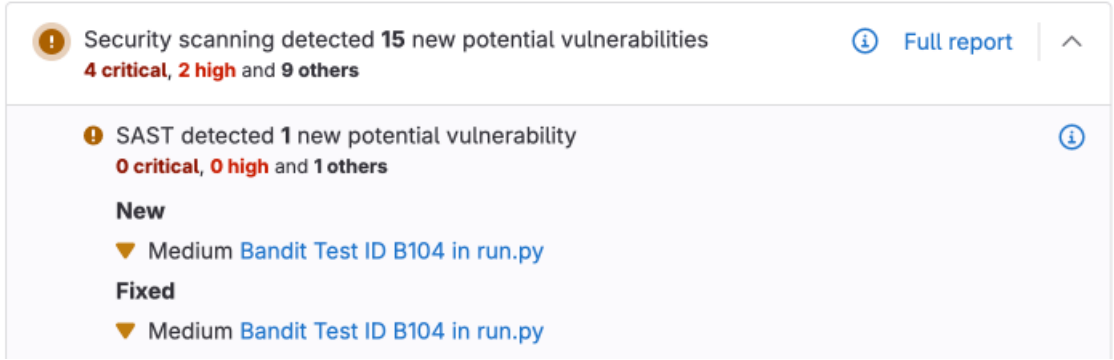
SAST

- What SAST misses
  - Cryptographic protocol misuse
    - Using crypto correctly per API but wrong for the threat model
  - Information disclosure through observable behavior
    - Timing attacks, error message variation
  - Anything requiring runtime context
    - Environment variables, deployed configuration, user roles in production



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CWE

- Common Weakness Enumeration (CWE)
  - CWE is a community-maintained taxonomy of software weaknesses
  - Every common vulnerability class has a CWE number
    - CWE-89 SQL Injection, CWE-79 XSS, CWE-798 Hardcoded Credentials, CWE-22 Path Traversal
  - Every modern SAST tool tags findings with CWE numbers
    - A finding tagged "CWE-89" means the same thing in Checkmarx, SonarQube, Snyk, Semgrep
  - The OWASP top 10 maps to CWE categories
    - The Top 10 is a curated list
    - The CWE is the vocabulary

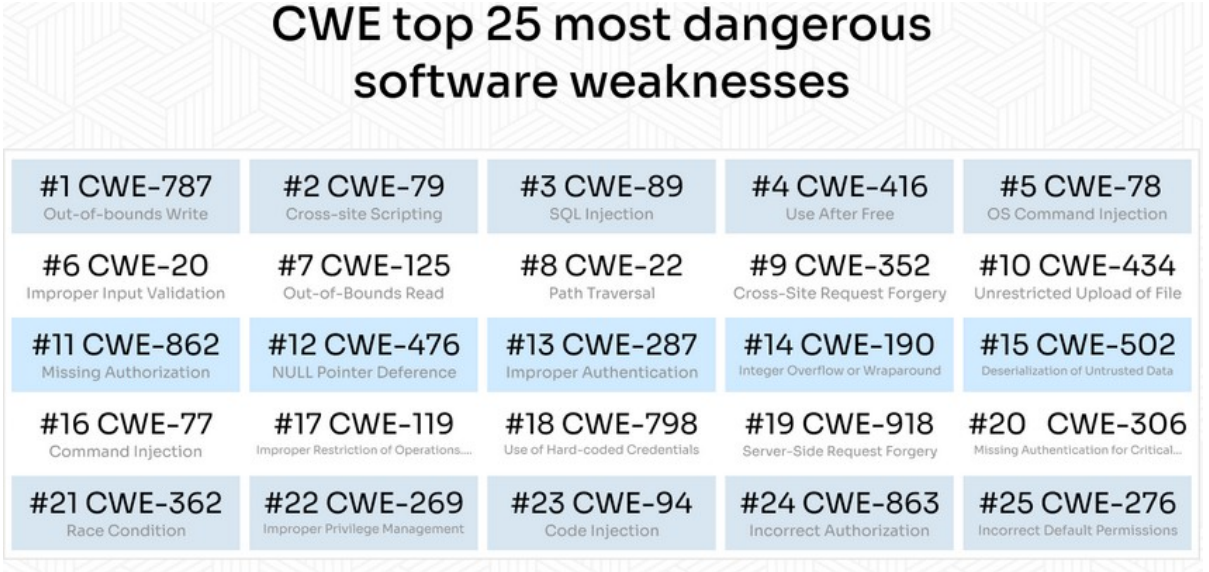
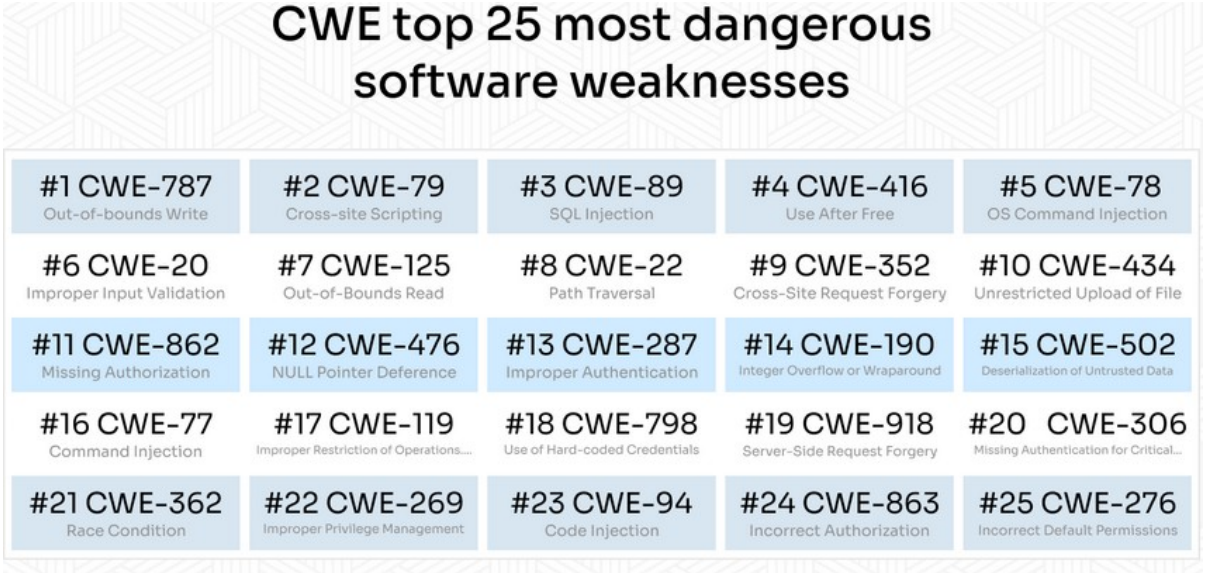


Image Credit: https://www.spinnakersupport.com/blog/2023/12/22/cve-vs-cwe/

CWE

- A few of the most common CWEs
  - CWE-89: SQL Injection
  - CWE-79: Cross-site Scripting (XSS)
  - CWE-78: OS Command Injection
  - CWE-22: Path Traversal
  - CWE-798: Use of Hard-coded Credentials
  - CWE-327: Use of a Broken or Risky Cryptographic Algorithm
  - CWE-352: Cross-Site Request Forgery (CSRF) -- this should ring a bell from Lab 4.8
  - CWE-502: Deserialization of Untrusted Data
  - CWE-200: Information Exposure



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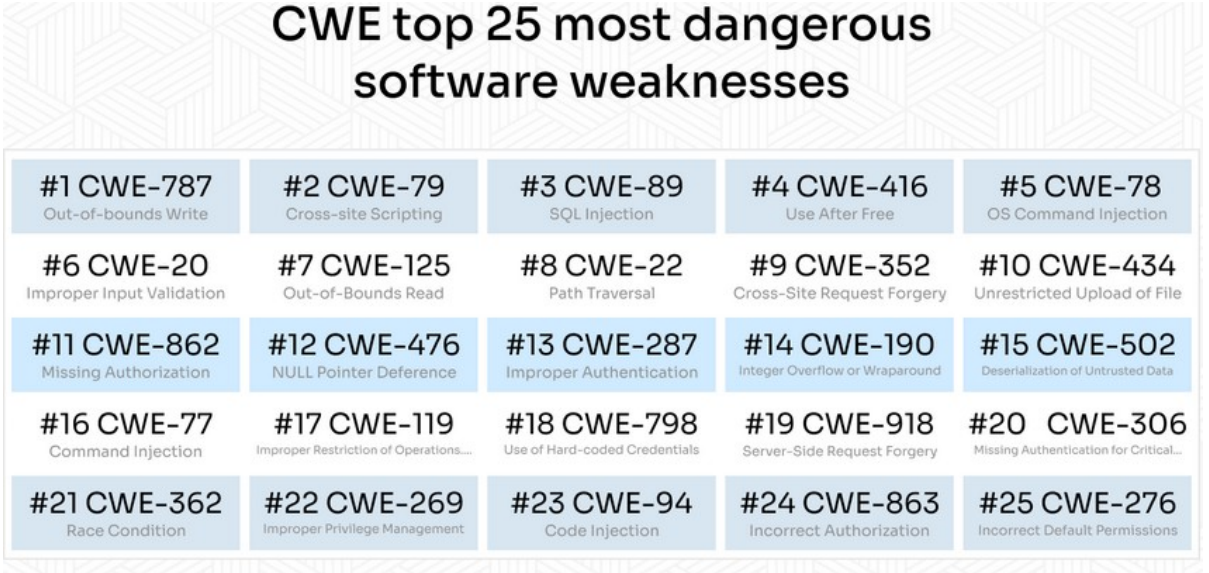
False Positives

- False positives
  - Every SAST tool reports things that aren't necessarily vulnerabilities
  - Because
    - A SAST tool that finds nothing is broken
    - A SAST tool that finds everything is useless
  - The tool errs on the side of reporting, leaves judgment to humans
  - "False positive" means the tool was wrong about the data flow or the risk



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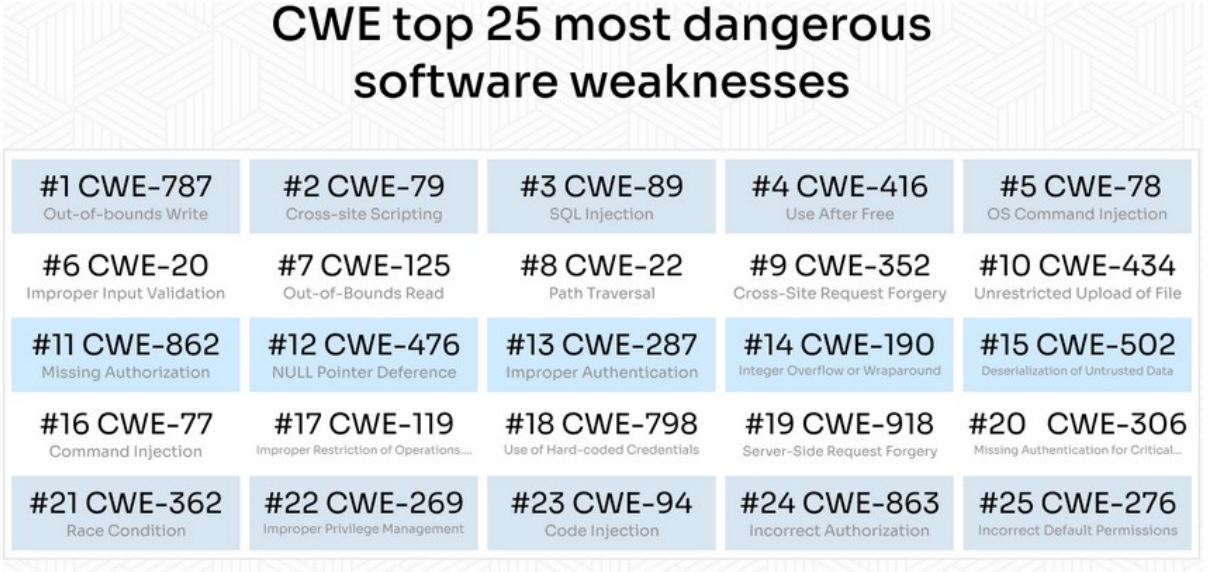
False Positives

- False positives
  - Every SAST tool produces them
    - This is a deliberate design choice
    - The tool defers judgment to a human (you) for the cases where it can't be certain
  - The skill is distinguishing real findings from false positives
  - Two general mistake types:
    - *Accept everything*: treat every finding as real, fix or suppress all of them, burn out on the volume
    - *Dismiss everything*: assume the tool doesn't understand and dismiss findings without analysis, miss real issues



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Triage Workflow

- Confirm
  - Read the data flow trace the tool provides
    - Most SAST tools show the path from source to sink, line by line.
  - Walk through it mentally
    - Is the data actually untrusted?
    - Does it actually reach the sink?
    - Is there a sanitizer the tool missed?
  - At the end of this step, you have a position
    - Yes, this is real
    - Or no, the tool got something wrong.

```
For each finding:

  1. CONFIRM      -- Read the data flow. Does the alleged vulnerability exist?

  2. CLASSIFY     -- Pick one:
                     - True positive, fix now
                     - True positive, accepted risk (document)
                     - False positive (document why)
                     - Out of scope (test code, generated, etc.)

  3. ACT          -- Fix the code OR document the suppression OR confirm scope

  4. VERIFY       -- Re-scan to confirm fix or that suppression applied correctly
```

Triage Workflow

- Classify
  - Four categories cover almost every finding
  - True positive, fix now
    - The vulnerability is real, has no compensating control, should be remediated immediately
    - SQL injection in user-facing code is the canonical example.
  - True positive, accepted risk
    - The vulnerability is real but mitigated by other factors
    - The endpoint is internal-only and behind authentication
    - The system is air-gapped
    - The compensating control is documented and reviewed by security
    - The classification acknowledges the issue while explaining why immediate fix isn't required.

```
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```

Triage Workflow

- Classify
  - False positive: the tool is wrong
    - Either the data flow doesn't actually exist (sanitizer missed)
    - Or the alleged risk doesn't apply (the sink isn't actually sensitive)
    - Document why so a future developer or auditor needs to understand the reasoning
  - Out of scope:
    - The finding is in code that doesn't get deployed
    - Test code, generated code, third-party libraries
    - Often these can be excluded by configuration before scanning, but if they sneak in, classify them as out of scope with a note.

```
For each finding:

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```

Triage Workflow

- Act
  - For "fix now," change the code
  - For "accepted risk" and "false positive," create the suppression in the tool with the documented justification
  - For "out of scope," either fix the scan configuration or mark them in the tool.
- Verify
  - Always re-scan after fixing
    - The finding should be gone (or at least transformed into a different finding).
    - If it's still there, your fix didn't address what the tool was complaining about.

```
For each finding:

  1. CONFIRM      -- Read the data flow. Does the alleged vulnerability exist?

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```

Example

- Triage
  - Read the trace
    - Does the user-supplied accountId actually reach the SQL execution unsanitized?
  - Inspect the code.
    - If the query uses + concatenation: it's real
    - If it uses ? placeholders with setString: false positive, then the tool missed the parameterization
  - Decide
    - True positive, fix now
    - False positive, suppress with justification.
  - Re-scan
    - Confirm the finding is gone after the fix

```
[HIGH] CWE-89 SQL Injection in TransactionRepository.java:42

Source:  request.getParameter("accountId") at TransactionController.java:18
Path:    accountId → transactionService.getByAccount(accountId) → ...
Sink:    statement.executeQuery(query) at TransactionRepository.java:42

Suggested fix: Use parameterized queries.
```

Severity Levels and Prioritization

- Not all findings are equal
  - Critical / High
    - Direct exploitability, business impact
    - RCE, SQL injection, hardcoded prod credentials
  - Medium
    - Exploitable in some contexts
    - XSS in admin pages, weak crypto for non-secret data
  - Low
    - Defense-in-depth issues
    - Verbose error messages, deprecated APIs, missing security headers
  - Info
    - Code quality observations, not exploitable on their own
  - CI pipeline typically blocks merge on Critical/High; tracks Medium; informational for Low

```
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Sink:    statement.executeQuery(query) at TransactionRepository.java:42

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```

Mapping SonarQube to Checkmarx

- Quick comparison
  - Cost
    - SonarQube Community is free and open source
    - Checkmarx is a commercial product with per-seat or per-scan licensing
  - Deployment
    - SonarQube Community runs locally in a Docker container
    - Checkmarx is typically a centrally-managed enterprise installation or SaaS subscription
  - Setup time for the labs
    - SonarQube takes minutes (one Docker command)
    - Checkmarx requires procurement, account provisioning, and security team coordination over weeks

Concept	SonarQube (lab)	Checkmarx (likely at the bank)
Severity	Blocker / Critical / Major / Minor / Info	Critical / High / Medium / Low / Information
Finding identity	Issue with key, rule, severity, location	Result with query name, severity, CWE
Triage status	Open / Confirmed / Resolved / Reopened	To Verify / Confirmed / Not Exploitable / Proposed Not Exploitable
False positive flag	Resolution: "False Positive" with comment	Mark as "Not Exploitable" with justification
Vulnerability vocab	CWE numbers	CWE numbers (same)
Scan trigger	Maven/Gradle plugin or CLI	CI integration, IDE plugin, CLI

Mapping SonarQube to Checkmarx

- Quick comparison
  - Java coverage
    - Both produce strong findings on Java code
    - The specific findings differ but the categories overlap substantially
  - Vulnerability vocabulary
    - Both use CWE numbers, so a finding's classification is portable between tools
  - Triage workflow
    - Both support the same workflow only the UI vocabulary differs
  - Banking adoption
    - Checkmarx is more common in large banks for the central CI pipeline
    - SonarQube is more common at the team/project level and for code quality analysis alongside security

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Suppression Justifications

- Writing Justifications for audit
  - Bad:
    - False positive."
    - "I don't think this is a problem."
  - Better:
    - "False positive. The tool identified accountId flowing to executeQuery without sanitization, but inspection shows the JDBC PreparedStatement uses positional parameters (line 42, statement.setString(1, accountId)).
    - The tool's data flow analysis did not recognize the parameterization through the prepareStatement / setString idiom. No remediation required."
  - Best:
    - [the better text above] +
    - "Reviewed with [security team contact] on [date]. Filed as suppression #1234 with reference to internal AppSec wiki page https://..."

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Suppression Justifications

- A bad justification ("false positive")
  - Tells the future reader nothing.
  - They have to redo your analysis from scratch, often without the context you had
  - They may incorrectly accept your dismissal
  - They may incorrectly reject it
  - Either way, you've left them worse off than necessary.

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Suppression Justifications

- A good justification has three properties:
  - States the tool's allegation
    - "The tool identified X flowing to Y."
    - This shows you understood what the tool was reporting, not just that you disagreed.
  - Explains the actual code behavior
    - "Inspection shows the code uses parameterized queries via PreparedStatement.setString."
    - This is the technical fact that contradicts the tool's allegation.
  - Identifies WHY the tool was wrong
    - "The tool's data flow analysis did not recognize the JDBC parameterization idiom."
    - This is the diagnostic insight that explains the false positive at the level of the tool's behavior, not just the code.

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Suppression Justifications

- The "Best" version
  - Adds review and traceability
    - A senior security person reviewed and concurred
    - A unique suppression ID in the tracking system links to the conversation
    - A wiki link or document reference provides extended context
    - This level of rigor is appropriate for high-severity findings or for codebases under regulatory scrutiny

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Dynamic Security Application Testing

DAST

- Dynamic application security testing
  - Dynamic
    - Runs against the actual running application
  - Black-box
    - The scanner does not see the source code, only the application's externally observable behavior
  - Late in the SDLC
    - Typically runs against staging or pre-production environments
    - The application is started
    - The scanner crawls it and discovers endpoints
    - Then probes those endpoints with crafted attack payloads
    - Observes responses to determine whether the application is vulnerable

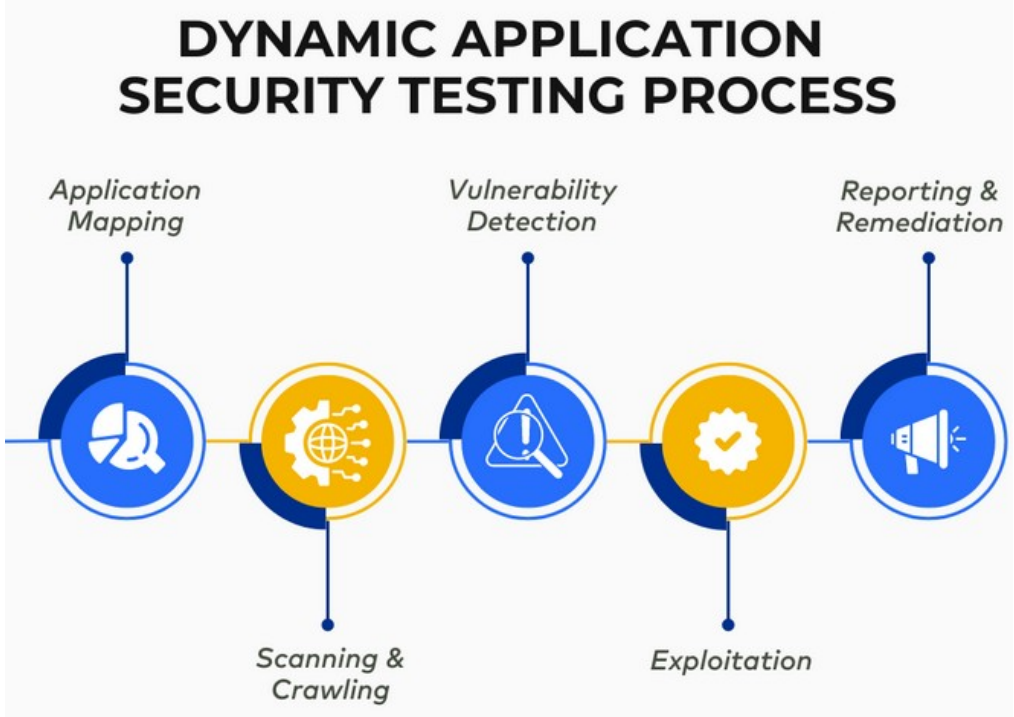

Image Credit: https://www.getastra.com/blog/dast/what-is-dast/

DAST Scan Lifecycle

- Lifecycle
  - Deploy
    - DAST requires a running application
    - Biggest practical constraint
    - Unlike SAST, DAST needs an environment to point at
    - Usually this is a staging environment, a deployed copy of the application that mirrors production.
  - Crawl
    - The scanner discovers what URLs the application has.
    - For traditional server-rendered apps with HTML links, this is straightforward
    - Follow every <a href>, submit every <form>
    - For modern SPAs like React crawling is harder because URLs are constructed by JavaScript at runtime.
    - Some DAST tools include a JavaScript engine to handle this
    - Others rely on developers providing a list of URLs to scan, often via a Postman/OpenAPI exports

```
1. DEPLOY     -- Application is running in staging environment
                 (real database, real config, real network)

2. CRAWL      -- Scanner discovers endpoints by following links,
                 submitting forms, exploring the URL space

3. PROBE      -- For each endpoint, send crafted attack payloads:
                 - "' OR 1=1--" for SQL injection
                 - "<script>alert(1)</script>" for XSS
                 - "../../../etc/passwd" for path traversal

4. OBSERVE    -- Watch responses for signs the attack succeeded:
                 - SQL errors in response
                 - Reflected payloads in HTML
                 - File contents leaking through

5. REPORT     -- Findings include the URL, the payload, the response,
                 and a CWE classification
```

DAST Scan Lifecycle

- Lifecycle
  - Probe
    - The scanner has a library of attack payloads tagged by vulnerability type
    - For each form field, query parameter, or header, it tries the relevant payloads
    - This is genuinely sending attack traffic to the application.
  - Observe
    - The scanner looks for signals that an attack worked
    - SQL injection signals: database error messages in the response.
    - XSS signals: the payload appearing in the response unescaped
    - Path traversal signals: file contents in the response
    - Each signal corresponds to a different finding type.

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DAST Scan Lifecycle

- Lifecycle
  - Report
    - Findings include the URL, the payload that triggered the finding, the response, and a CWE classification
    - Same CWE vocabulary as SAST.
  - DAST scans are slow
    - A thorough DAST scan against a moderately-sized web application can take hours
    - They also generate noise in production logs
    - This is why DAST runs in staging, not production.

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DAST and SAST

- What DAST does that SAST doesn't
  - Runtime configuration issues
    - Environment variables
    - Deployed application.yml values
    - Infrastructure settings
  - Deployment misconfigurations
    - Exposed admin endpoints
    - Debug mode enabled
    - Default credentials
  - Authentication flow problems
    - Session fixation
    - Token replay
    - password reset bypasses

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DAST and SAST

- What DAST does that SAST doesn’t
  - Server-level vulnerabilities
    - Web server misconfigurations
    - Missing security headers
    - Weak TLS
  - Real-world exploitability
  - The SAST tool says "this could be exploited";
  - DAST proves "this WAS exploited"

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DAST and SAST

- What SAST does that DAST doesn’t
  - Code paths the scanner cannot reach
    - Internal APIs
    - Admin endpoints
    - Code behind feature flags
    - Error paths
  - Code quality issues that aren't yet exploitable
    - Weak crypto in development
    - Hardcoded credentials in source
    - Deprecated APIs

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DAST and SAST

- What SAST does that DAST doesn’t
  - Vulnerabilities in
    - Non-exposed code
    - Internal microservice APIs
    - Batch jobs and scheduled tasks
  - Per-finding precision
    - SAST points at the exact line
    - DAST points at the URL
  - Coverage of cold paths
    - Code that runs rarely like error handlers, infrequent business cases
    - Gets analyzed by SAST regardless of whether it runs or not

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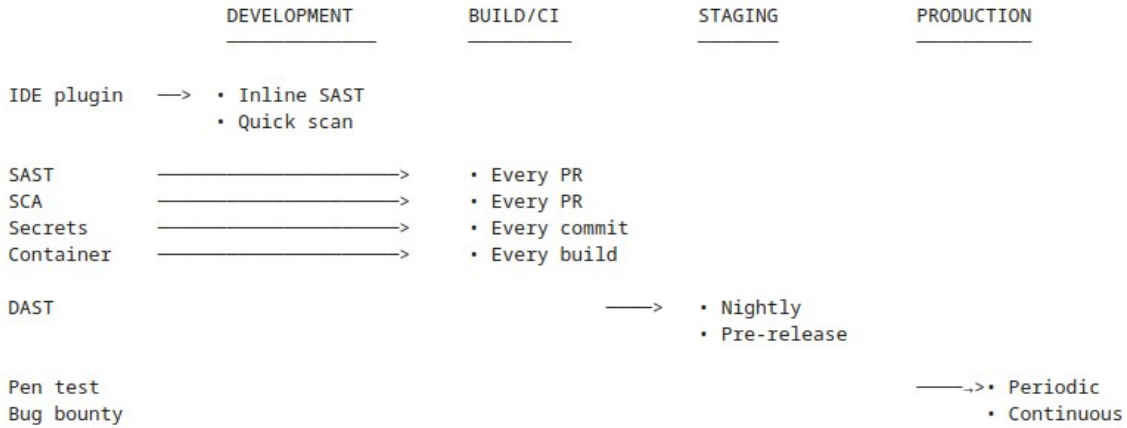
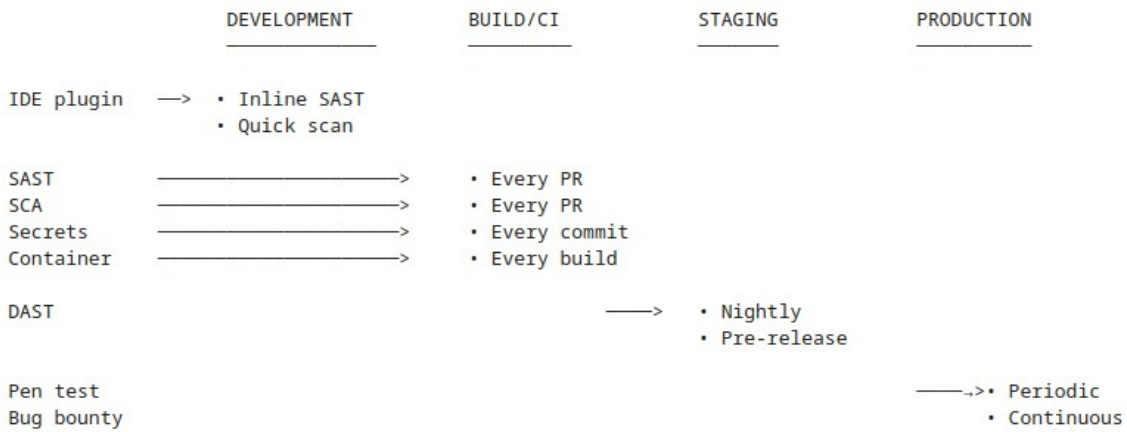
SDLC Fit

- For the bank context
  - All these layers exist in mature programs.
- Banks typically have
  - SAST and SCA in CI
    - For the bankbff, this would be triggered on every push
  - DAST runs against staging environments before production releases
  - Penetration tests at least annually, often more often for high-value applications
  - A bug bounty or vulnerability disclosure program for external reporting



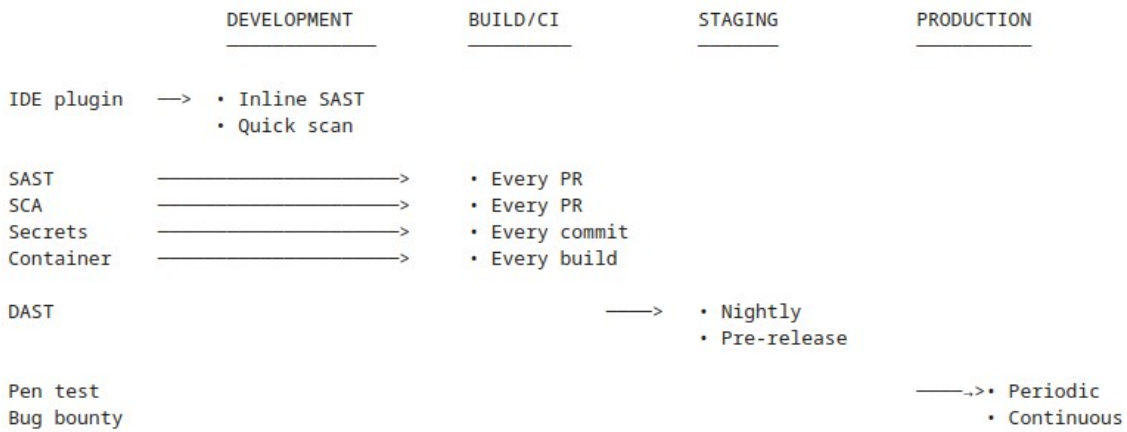
Other AppSec Layers

- Beyond SAST and DAST
  - SCA (Software Composition Analysis)
    - Scan dependencies for known vulnerabilities (CVEs)
  - Secrets scanning
    - Detect API keys, passwords, certificates accidentally committed to source control
  - Container scanning
    - Analyze Docker images for vulnerable OS packages and configurations
  - IAST (Interactive Application Security Testing)
    - Runtime instrumentation that combines SAST and DAST signals
  - Penetration testing
    - Humans, manually, attacking the application to find what scanners cannot
  - Bug bounty
    - External researchers paid to find and report vulnerabilities



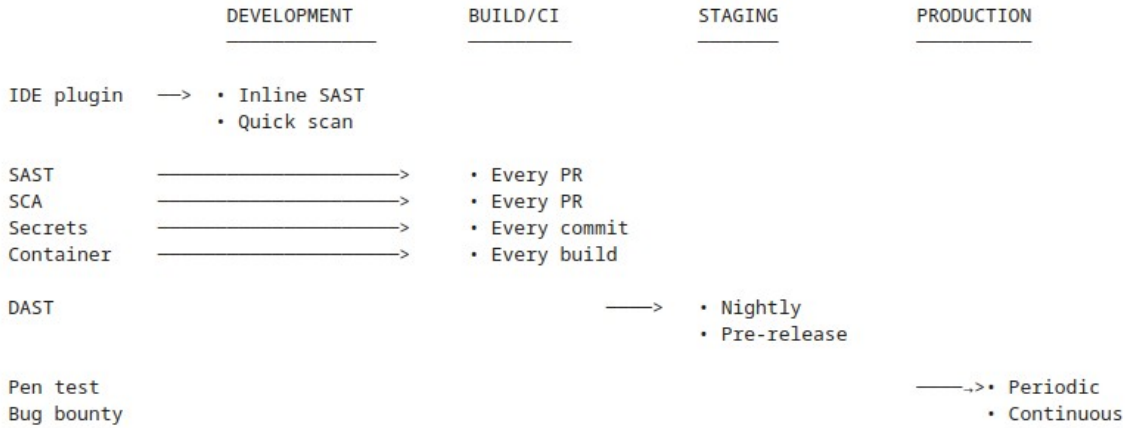
The Developer's Role

- You own
  - SAST findings
  - SCA findings
  - Secrets scanning findings
  - Fixing the vulnerabilities those tools surface
- You partially own
  - Unit and integration tests
  - Tests that catch security regressions



The Developer's Role

- Security engineers own
  - Running DAST scans
  - Configuring tooling
  - Maintaining baselines
  - Escalating findings
- Pen testers own
  - Finding what tools cannot find; their reports become work items for developers
- Bug bounty support own
  - Validating external reports
  - Valid findings become work items for developers



Questions?